LECTURE 15

LAST TIME

LECTURE 15: OUTLINE

Greeks

THIS TIME

LECTURE 16: OUTLINE

Trading volatility (subtext: Greeks not independent)

Implied volatility

TRADING VOL

INSTANTANEOUS VOL

Recall σ defined as $\sigma^2 t \equiv \mathrm{Var}^{\mathbb{Q}}\left[\log \frac{S_t}{S_0}\right]$

"Instantaneous vol" σ: over small time increment h

$$\sigma^2 h \equiv \mathrm{Var}^{\mathbb{Q}}\left[\log \frac{S_{t+h}}{S_t}\right] \to \mathrm{Var}^{\mathbb{Q}}\left[\frac{S_{t+h}}{S_t}\right] \text{ (since } \log(1+x) = x - \tfrac{1}{2}x^2 + \cdots\text{)}$$

Instantaneous vol constant σ in B–S: worst assumption of B–S...

VOL TO MATURITY

But all that matters is B–S is the "vol to maturity" $\sigma\sqrt{T-t}$

If vol not constant but deterministic, can do B–S with vol to mat.

$$\bar{\sigma}\sqrt{T-t} := \sqrt{\int_t^T \sigma_s^2 ds}$$

But non-deterministic vol is a problem for B–S

HOW CAN YOU TRADE VOL?

To long vol, use vol. pump: Long call (convex); short rep. port. (linear)

Discrete rebalancing: $\text{P\&L}_t \approx \frac{\partial V_t}{\partial t}\Delta t + \frac{\partial V_t}{\partial S}\Delta S + \frac{1}{2}\frac{\partial^2 V_t}{\partial S^2}(\Delta S_t)^2$

NB: $\Delta S >> \Delta t$, so other second-order terms are about zero

Use
$\mathbb{E}^{\mathbb{Q}}[\text{P\&L}] \approx rV\Delta t,\ \mathbb{E}^{\mathbb{Q}}[\Delta S] \approx rS\Delta t,\ \mathbb{E}^{\mathbb{Q}}[(\Delta S)^2] \approx \text{Var}^{\mathbb{Q}}[\Delta S] = \sigma_t^2\Delta t$:

$$rV = \frac{\partial V_t}{\partial t} + rS\frac{\partial V_t}{\partial S} + \frac{1}{2}\sigma_t^2\frac{\partial^2 V_t}{\partial S^2}$$

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$$rV = \frac{\partial V_t}{\partial t} + rS\frac{\partial V_t}{\partial S} + \frac{1}{2}\sigma_t^2\frac{\partial^2 V_t}{\partial S^2}$$

Vol pump: $\Pi = c - \beta B - \Delta S \implies \Theta = -\frac{1}{2}\sigma_t^2\Gamma$

GAMMA RENT VS. THETA BLEED

Use $\Theta = -\frac{1}{2}\sigma_t^2\Gamma$ and $\frac{\partial V_t}{\partial t} = -\frac{1}{2}\sigma_t^2\frac{\partial^2 V_t}{\partial S^2}$ to write realized P&L:

Zero value $V_t = 0$ and delta hedge $\partial V/\partial S = 0$ imply

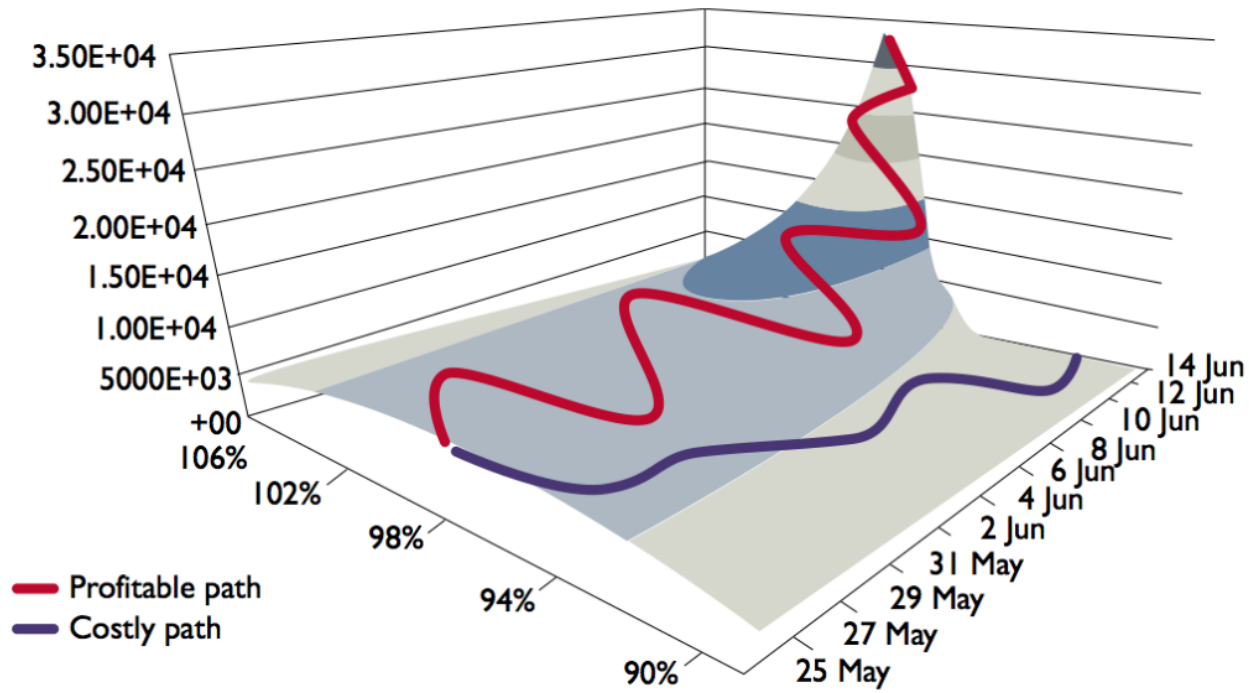
$$\text{P\&L} \approx \frac{1}{2}\sum_t \Big(\Delta S_t^2 - \sigma_t^2\Delta t\Big)\frac{\partial^2 V_t}{\partial S^2} = \frac{1}{2}\sum_t \Big(\Delta S_t^2 - \sigma_t^2\Delta t\Big)\Gamma_t$$

ΔS_t^2 like "realized variance" and $\sigma_t^2\Delta t$ like "implied variance"

Can P&L < 0 if realized variance higher than implied on average?

GAMMA PATHS (BLEEDING TO DEATH)

CHART 4: RWE 06/06 C 74 – GAMMA FOR 1000 LOTS (DATE = PRICING DATE, VERTICAL LEFT HAND AXIS IS GAMMA)



Source: ABC Square Asset Management LLP/Fimat Alternative Investment Solutions

IMPLIED VOL

BLACK-SCHOLES AND VOL

B–S price depends on log-normal distribution of S_T

Normality due to assumption of const. vol (const. u and d on tree)

In reality, vol changes

Changes in S_t are correlated in time

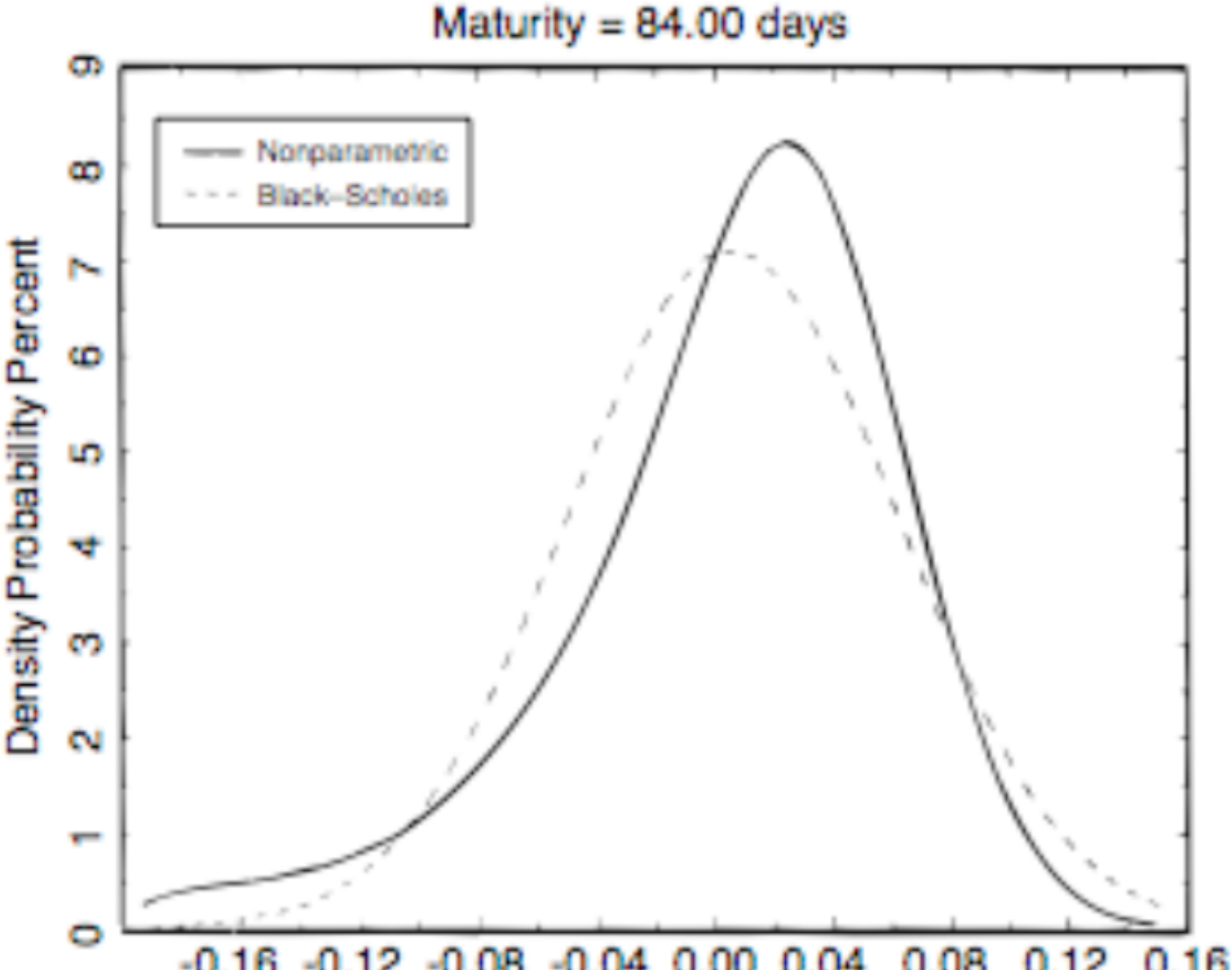
Volatility clusters

$\Longrightarrow$ fat tails

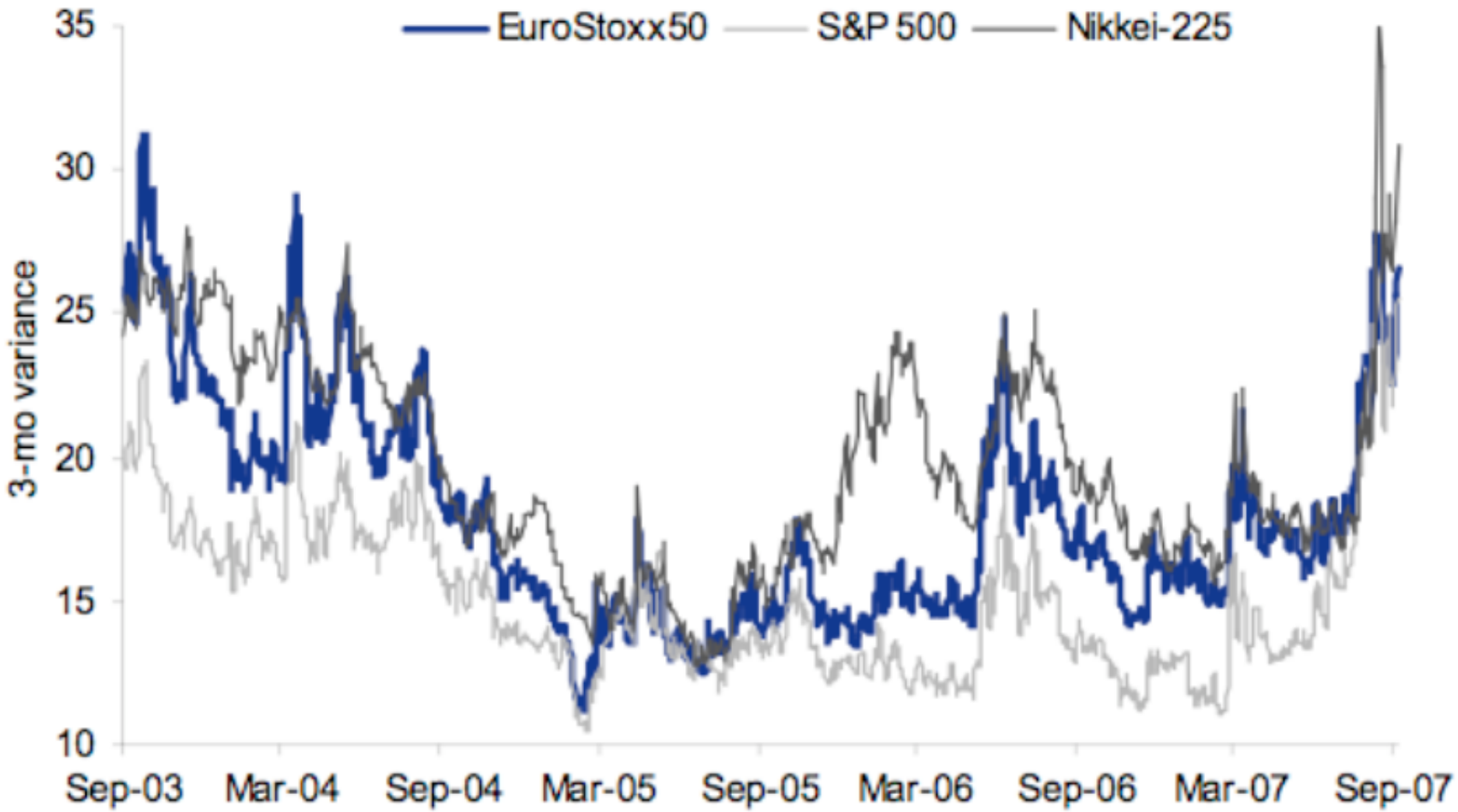
DISASTERS ARE COMMON

Date	**Return (%)**	z**-score** ($\sigma = 15\%$)	**Once in (years)**
1987-10-19	-20.47	-21.6	10^{101}
1929-10-28	-12.34	-13.0	10^{36}
2020-03-16	-11.98	-12.7	10^{34}
1929-10-29	-10.16	-10.7	10^{23}
1929-11-06	-9.92	-10.5	10^{22}
2020-03-12	-9.51	-10.0	10^{21}
1937-10-18	-9.27	-9.8	10^{20}
2008-10-15	-9.04	-9.6	10^{19}
2008-12-01	-8.93	-9.4	10^{18}
1933-07-20	-8.88	-9.4	10^{18}
2008-09-29	-8.79	-9.3	10^{18}
1933-07-21	-8.70	-9.2	10^{17}
1987-10-26	-8.28	-8.8	10^{14}
1932-10-05	-8.20	-8.7	10^{14}
1932-08-12	-8.02	-8.5	10^{14}
Age of the Earth			4.5×10^9
Age of the Sun			4.6×10^9
Age of the Milky Way			1.3×10^{10}
Age of the Universe			1.38×10^{10}

FAT TAIL


Maturity = 84.00 days
Nonparametric
Black-Scholes
Density Probability Percent
0
1
2
3
4
5
6
7
8
9
-0.16
-0.12
-0.08
-0.04
0.00
0.04
0.08
0.12
0.16

VOL NOT CONSTANT, UNLIKE IN B–S



Source: Goldman Sachs.

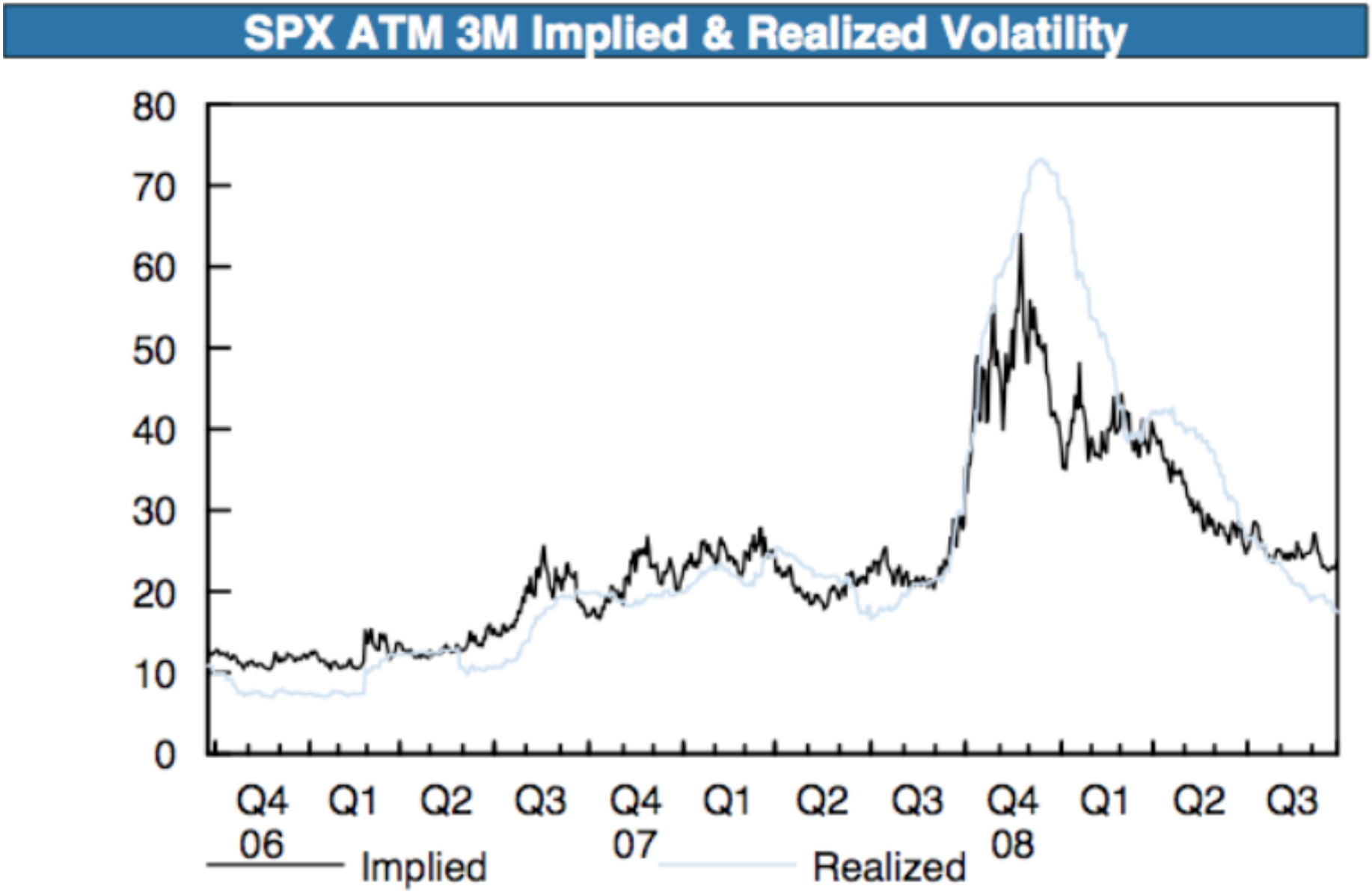
B–S VS. MARKET PRICES

Can make B–S price c^{BS} to equal market price c^* for some σ_I

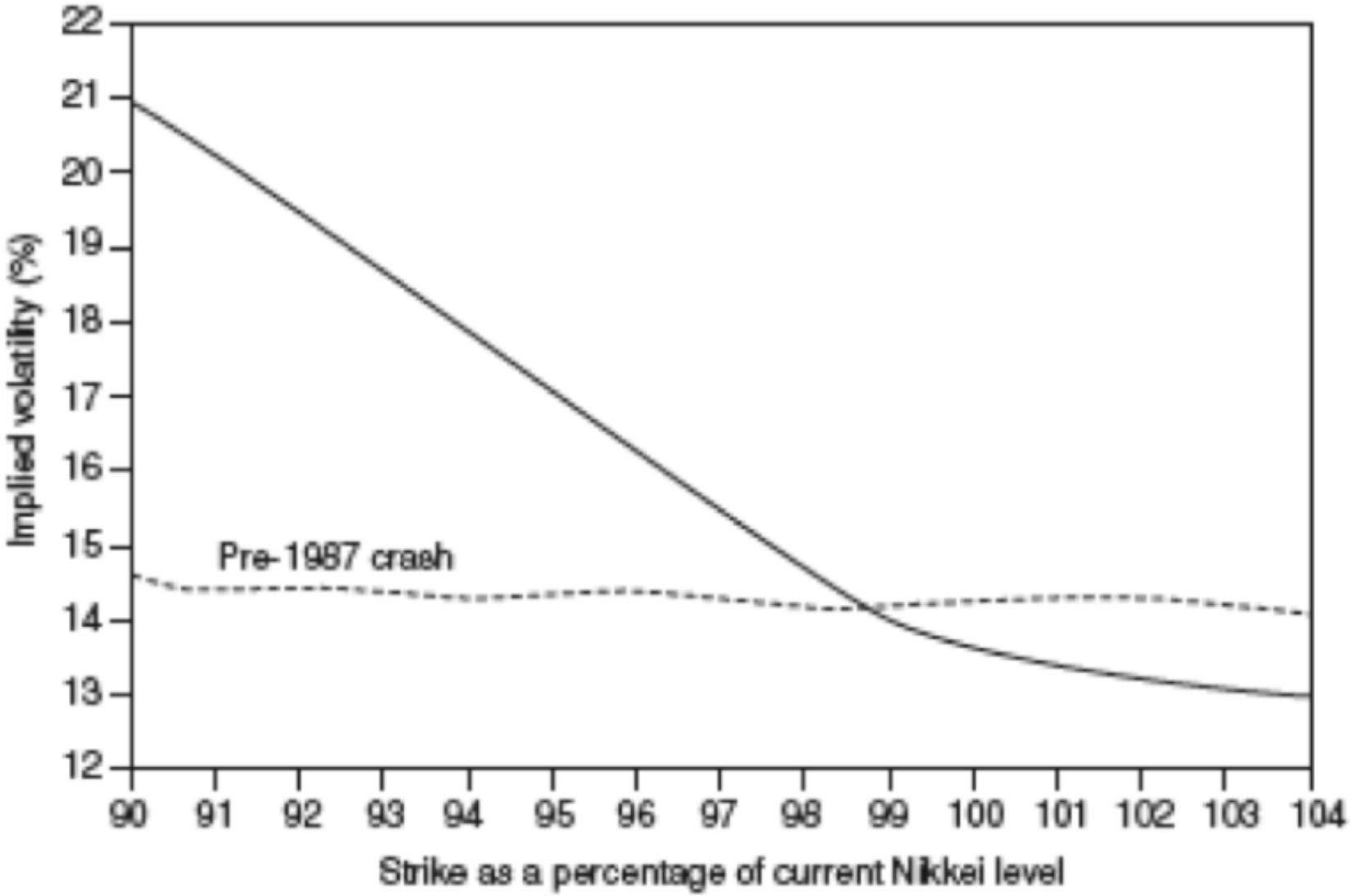
$$c^*\big(t, S, K, T, r\big) = c^{\mathrm{BS}}\big(t, S, K, T, \sigma_I\big)$$

Suppose B–S assumptions. Does implied vol σ_I change?

IMPLIED VOL ≠ REALIZED VOL



VOL SMIRK


Implied volatility (%)
22
21
20
19
18
17
16
15
14
13
12
Pre-1987 crash
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
Strike as a percentage of current Nikkei level

FAT LEFT TAIL

Vol smirk implies that o.t.m. puts have high implied vol

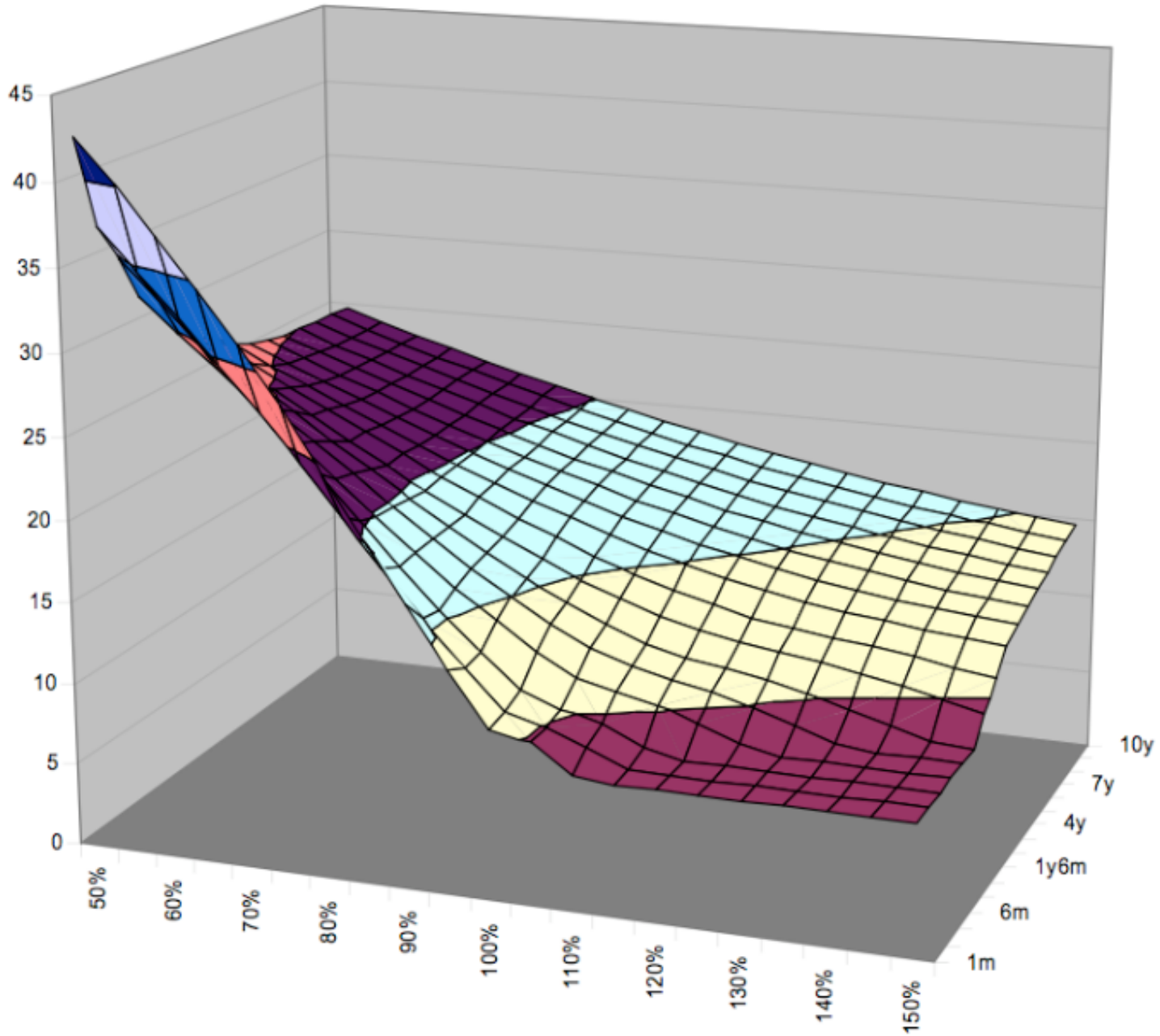
High implied vol = high price ($\nu > 0$)

Risk-neutral prob of large price falls higher than normal

I.e. there is a fat left tail

Right tail is a bit fat too, hence sometimes called smile not smirk

SKEW FLATTENS OUT WITH LONGER TTM



RECAP

LECTURE 15: RECAP

Volatility is not constant

Clustering and fat tails

FIN